

CTC + OT for ST

Pre-training for Speech Translation

Authors: Phuong-Hang Le¹, Hongyu Gong², Changhan Wang², Juan Pino², Benjamin Lecouteux¹, Didier Schwab¹

Affiliations: ¹ Univ. Grenoble Alpes, CNRS, Grenoble INP, LIG, France • ² Meta AI

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Code: github.com/formiel/fairseq

Contact: hang.le@univ-grenoble-alpes.fr

Key Ideas

- Modality gap in ST: speech encoder pre-trained on audio vs. text decoder pre-trained on text.
- CTC pre-training reduces the gap by learning alignments without a decoder.
- Propose Siamese-like pre-training: speech and text encoders aligned via Optimal Transport (Wasserstein distance).
- Positional encoding in OT enforces monotonic alignment.
- State-of-the-art under no-external-data; competitive with strong multi-task systems; stackable on them.

Intro and Gap

- End-to-end ST often pre-trains ASR (speech) and MT (text) components.
- Standard recipe: ASR(CE) encoder + MT(CE) decoder → initialize ST.
- Decoder/encoder used for alignment are later discarded → alignment info lost.
- Result: pre-train / fine-tune mismatch.

CTC Basics

- CTC predicts frame-level tokens; COLLAPSE removes repeats via blanks; Viterbi sums alignments.
- Encoder learns alignment without decoder → preserves needed info for ST.
- CE can yield stronger ASR with decoder, but CE-encoder alone weaker for ST.
- CTC is partial: does not recover loss from removing MT encoder.

Related Work

- ST: From cascaded ASR → MT to competitive end-to-end models.
- Bridging gap: cascaded encoders, adaptors, adversarial regularizers, multi-task.
- CTC in pre-training: used but less quantified vs. CE; we provide analysis.
- Speech-text alignment: Euclidean/cosine/KL/contrastive; OT naturally handles length mismatch.

OT Primer

- Goal: transport mass from distribution α to β with minimal cost.
- Wasserstein distance W ; use entropic-regularized W_λ (Sinkhorn) for efficiency.
- Handles sequences of different lengths $n \neq m$ naturally.

Loss $\approx W_\lambda(\alpha, \beta)$ with cost $c(u_i, v_j) = ||u_i - v_j||_p$; differentiable and efficient on GPU.

Siamese OT Pretraining

- Two encoders: speech and text; align outputs via $W_\lambda(U, V)$.
- Joint loss: CTC (speech) + $\alpha \cdot \text{OT}(U, V)$.
- Addresses speech–text representation mismatch at pre-train stage.

Positional OT

- Monotonicity prior: early frames $\leftrightarrow$ early tokens.
- Append normalized positions to features: $u'=[u; y_s]$, $v'=[v; y_t]$.
- Makes cost order-aware while preserving OT efficiency.

$$C_{ij} = ||[u_i; y_{s_i}] - [v_j; y_{t_j}]||_p$$

ST Pretraining

- With MT pre-train: init text encoder from MT encoder during ASR Siamese-PT.
- Without MT: randomly init text encoder; still effective.
- ST fine-tune: init with ASR encoder + MT decoder.

Experimental Setup

- Datasets: MuST-C (En → 8 EU langs), CoVoST-2 (multilingual; many low-resource).
- Models: Transformer; speech enc 12L, speech dec 6L, text dec 6L; d=512/1024.
- Losses: CTC + α ·(CE or OT), α =0.1; OT: λ =1, ℓ_2 cost; positional weight γ =1.0.
- Evaluate on dev for ablations; test for SOTA comparison.

Qualitative Successes

Example A (MuST-C En → De)

Baseline (CTC): misorders adjective–noun pair. With OT: corrects monotonic alignment, yields proper word order.

Callout: OT fixes word order aligned with speech timing.

Example B (CoVoST-2 En → Fr)

Baseline (CTC): drops short function word. With OT: preserves token due to better frame–token matching.

Callout: OT reduces deletion errors under length mismatch.

More Examples

Example C (En → Es)

Baseline (CE pre-train): number mistranslated. CTC pre-train: better alignment, correct numeral.

Callout: CTC alignment improves number fidelity.

Example D (En → It)

CTC alone: partial phrase repetition. CTC+OT: removes repetition via consistent alignment map.

Callout: OT curbs repetition from mis-collapsed frames.

Failure Cases

Example E (Very low-resource $X \rightarrow \text{En}$)

CTC+OT still produces inaccurate rare named entities; variance high on tiny splits.

Callout: Limited data $\rightarrow$ unstable alignments; entities remain wrong.

Example F (Long utterance, $\text{En} \rightarrow \text{Ru}$)

OT improves local order but misses long-distance agreement; trailing clause omitted.

Callout: Long sequences strain alignment; omissions persist.

Results and Takeaways

- Dev (avg): CTC > CE; CTC+OT best across MuST-C and CoVoST-2 (excl. ultra low-resource).
- MuST-C test (large, multi, no ext): CE 28.8, CTC 29.2, CTC+CE 29.1, Siamese-PT 29.8 BLEU.
- CoVoST-2 test (no ext): $\text{En} \rightarrow X$ 21.5 (Siamese-PT), $X^{**} \rightarrow \text{En}$ 25.5; improves over prior baselines.
- Multi-task add-on ($\text{En} \text{--} \text{De}$): CE 26.78 $\rightarrow$ CTC 27.04 $\rightarrow$ Siamese-PT 27.20.
- Limits/Future: very low-resource remains hard; explore external data, tune λ , γ ; OT as auxiliary loss.

Closing Notes

- Ethics: potential surveillance use; apply privacy safeguards.
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